

Classless Inter-Domain Routing (CIDR) & IP Addressing Mechanics

Summary

Classless Inter-Domain Routing (CIDR) replaced rigid classful addressing to eliminate massive IP wastage and improve routing scalability. Instead of fixed Class A/B/C blocks, CIDR treats the entire (2^{32}) IPv4 space as a flexible pool and allocates address blocks based on actual need.

The core technical principles are:

- Variable-length prefix notation (**a.b.c.d/n**)
- Power-of-two block sizes
- Contiguous allocation
- Divisibility rule for valid block alignment
- Efficient route aggregation using longest-prefix logic

Mastering these rules makes CIDR problems mechanical rather than conceptual.

1 Evolution: From Classful to Classless

Limitations of Classful Addressing

Classes	Addresses per Network	Problem
A	$(2^{24}) = 16,777,216$	Massive wastage
B	$(2^{16}) = 65,536$	Often too large
C	$(2^8) = 256$	Often too small

Example:

If an organization needed (2^{22}) addresses, it had to take a Class A block → wasting over 12 million addresses.

Subnetting: A Temporary Fix

Subnetting allowed internal division of classful blocks but:

- Still depended on rigid class allocation
- Could not globally optimize distribution
- Did not prevent large-scale wastage

Hence the need for CIDR.

2 Foundations of CIDR

CIDR removes classes entirely.

Instead of fixed sizes, we use:

a.b.c.d / n

Where:

- **n** = prefix length (network bits)
- **32 - n** = host bits
- Total addresses = $(2^{\{(32-n)\}})$

CIDR Structure

Component	Meaning
Prefix (/n)	Network ID length
Suffix	Host ID
Total IPs	$(2^{\{(32-n)\}})$

Usable Hosts $(2^{\{(32-n)\}} - 2)$

3 Block Calculation Mechanics

Finding Block ID (First Address)

1. Keep first n bits unchanged
2. Set remaining $(32-n)$ bits to 0

This gives the Network/Block ID

Finding Direct Broadcast Address (Last Address)

1. Keep first n bits unchanged
2. Set remaining bits to 1

This gives the DBA

4 The Three Mandatory CIDR Rules

These rules ensure routing correctness and aggregation efficiency.

Rule 1: Contiguous Addresses

All IPs in the block must be consecutive.

No jumps.

No fragmentation.

Rule 2: Block Size Must Be Power of Two

[
 $\text{Block Size} = 2^k$
]

Why?

Because binary division by (2^k):

- Last k bits = Host ID
- Remaining bits = Network ID

This allows instant hardware separation of network and host components.

Rule 3: Divisibility Rule

The first IP address must be divisible by the block size.

[
 $\text{First IP} \bmod 2^k = 0$
]

Why?

If divisible by (2^k):

- The last k bits are zero
- That automatically makes it a valid Block ID

If this condition holds:

- The last address will automatically have all 1s in the Host portion
- It becomes a valid DBA

5 Practical Examples

✓ Valid Block Example

Range:

100.100.100.64 – 100.100.100.127

Step 1: Contiguous?

Yes (64 → 127)

Step 2: Size?

[
127 - 64 + 1 = 64 = 2⁶
]

Step 3: Divisible?

64 ÷ 64 = 1 ✓

Valid block.

CIDR notation:

100.100.100.64 / 26

✗ Invalid Block Example

Range:

100.100.100.1 – 100.100.100.32

Size = 32 = (2⁵) ✓

Contiguous ✓

Check divisibility:

1 ÷ 32 → Not divisible ✗

Invalid block because:

- Host bits of first address are not zero.

6 Full Derivation Example

Given:

100.100.100.68 / 27

Step 1: Determine Host Bits

[
32 - 27 = 5
]

Total IPs:

[
 $2^5 = 32$
]

Step 2: Identify Block Boundary

Block size = 32

Find nearest multiple of 32 $\leq$ 68

Multiples:

0, 32, 64, 96...

Nearest lower multiple = 64

Step 3: Final Block Details

Parameter	Value
Block ID	100.100.100.6 4
First Host	100.100.100.6 5

Last Host 100.100.100.9
4

DBA 100.100.100.9
5

Total IPs 32

Range:

100.100.100.64 – 100.100.100.95

7 Why CIDR Is Powerful

1. Eliminates Class Boundaries

Any prefix length is allowed (/13, /19, /27, etc.)

2. Enables Route Aggregation

Multiple small networks can be summarized into one larger prefix.

3. Reduces Routing Table Size

Core routers depend heavily on prefix aggregation.

8 Longest Prefix Match (LPM) Connection

CIDR works with routing via:

If multiple routes match, choose the route with the largest prefix (most bits).

This ensures:

- More specific routes override general ones
- Aggregation does not break routing correctness

